Right-Sized Intelligence: Using Model Capacity and Training Stage to Choose Effective Large Language Models (LLMs)

A Descriptive Analysis of Capacity, Training Stage, and Performance in Open Weight LLMs

Introduction

Artificial intelligence is reshaping industries at a rapid pace, yet organizations tackling humanity's most urgent challenges risk being priced out of the revolution. For budget-constrained non-profits, Large Language Models (LLMs) offer a promising way to expand capacity, but these organizations typically cannot afford premium API agreements for frontier models or the compute infrastructure to host massive open-weight models locally. Many turn to smaller, efficient models from platforms like Hugging Face, only to face a sprawling ecosystem with limited time or engineering capacity to evaluate which models fit their hardware constraints while maintaining acceptable accuracy.

This study addresses that challenge by conducting a statistical analysis of Hugging Face LLMs. By evaluating the tradeoffs between model footprint and output quality, we seek to answer a key question: **Which open-source LLMs provide an effective balance of task performance and low computational overhead, enabling resource-constrained organizations to deploy practical AI tools?**

Our analysis reveals that simply choosing a larger model is not a guaranteed path to better performance; the model's training stage fundamentally alters the return on investment for scaling. Specifically, we identified a performance sweet spot at the 10-billion parameter mark, where models deliver above average performance and can be run locally on a laptop. At this scale, Base-Merge models outperform models at other training stages. For organizations with even stricter hardware limitations (under 5 billion parameters), standard Pretrained models offer the most efficient baseline performance.

Data and Methods

Data Source and Sample Construction

We used data from the public Open LLM Leaderboard "contents" dataset hosted by Hugging Face. The leaderboard provides standardized evaluation scores and metadata for many open weight large language models. Our extract contains 4,576 observations.

Following standard data cleaning practices, we standardized variable names, removed missing values, flagged models that were taken down for re-evaluation, and dropped models with non-positive parameter counts. To mitigate potential violations of the Independent and Identically Distributed (IID) assumption we deduplicated models with multiple precision values, and removed models prior to a June 26, 2024 Leaderboard overhaul. After applying these criteria, our final sample size is 4,285.

Principal Component Analysis

To evaluate the potentially overlapping leaderboard benchmarks, we conducted a principal component analysis (PCA). Our dataset satisfies all necessary assumptions for PCA aside from having outliers. The 40 outliers were all high-performing models that are relevant to our research question, and PCA results were not materially different without them so we retained them in the analysis. The assumptions were supported by the correlation matrix in Figure 1 (link) and Kaiser-Meyer-Olkin Test scores (0.81-0.94). The loading matrix in Table 1 (link) shows the proportion of variance accounted for by PC1 and PC2 as well as how much each benchmark contributes. PC1 accounts for a substantial 73% of variance and PC2 12%, together covering 85% of benchmark variance. PC1's high loadings across benchmarks (0.32-0.45) suggest it is a strong single summary of general performance. Within PC1, MMLU-Pro (benchmark asking challenging questions across diverse disciplines) has the highest loading and therefore serves as the best proxy for overall capability. Instruction-following benchmark IFEval's strong representation in PC2 and contrast with MUSR (long-range soft reasoning across narratives) and GPQA (graduate level sciences) demonstrate a tradeoff between instruction-following and reasoning. These results are seen in Figure 2 via a biplot colored by squared cosine. Capturing information from PC1 and PC2 with MMLU-Pro and IFEval as proxies addresses multicollinearity and helps reduce overfitting, while keeping coefficients interpretable.

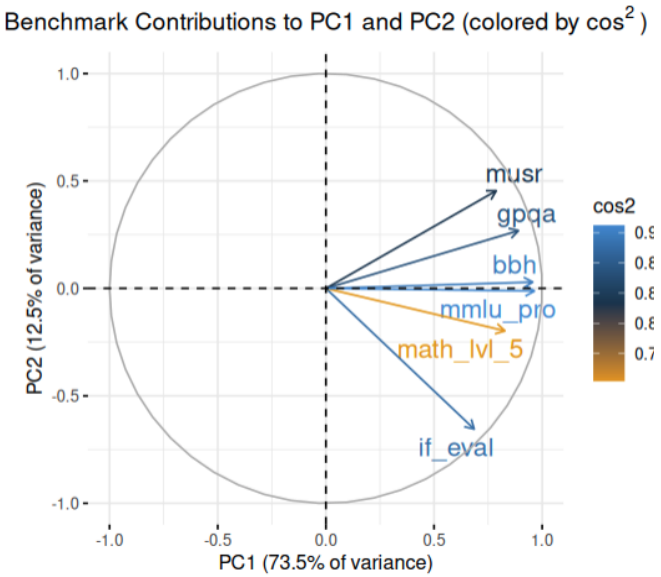


Figure 2: Biplot of Benchmark Contributions and Representation in PCs

Empirical Strategy

Given the relatively small dataset, we did not split it for separate exploration and confirmation. To mitigate our knowledge of the data leading to deceptively small p-values we constrained our tests. We framed a specific research question upfront and before starting wrote down one final model with the 4 iterations leading to it.

Our approach for this analysis is grounded in the foundational scaling laws for neural language models, which establish a predictable power law relationship between a model's parameter count and its training loss (Kaplan et al., 2020; Hoffmann et al., 2022). This implies a log-linear relationship between parameter count and benchmark performance, which the shape of the data matches. The performance, measured via the MMLU-Pro benchmark as the strongest

representation of general performance, is modeled on the base-10 logarithm of the model's parameter count and its training configuration with instruction-following ability controlled.

$$\text{MMLU-Pro}_i = \beta_0 + \beta_1 \log_{10}(\text{Size}_i) + \beta_2 \text{TrainingStage}_i + \beta_3 (\log_{10}(\text{Size}_i) \times \text{TrainingStage}_i) + \beta_4 \text{IFEval}_i + \epsilon_i$$

The primary independent variable, number of parameters, is log10 transformed to account for diminishing marginal returns in model scaling. The transformation linearizes this relationship. Consequently, our coefficient can be interpreted as the expected change in the MMLU-Pro score associated with a tenfold percentage increase in model size.

Training stage categories are defined as:

- **Pretrained:** raw base model
- **Fine-Tuned:** domain-specific data added to Pretrained
- **Base-Merge:** models fused together
- **Chat-Instruct:** base models tuned for interaction
- **Other (<60 models):**
  - Multimodal: trained with text + images/video/audio
  - Continuously Pretrained: base model further trained

The coefficient on these indicators reflects the incremental performance associated with specific post-training strategies relative to a Base-Merge model, assuming a constant parameter count.

To isolate the general performance from pure formatting compliance, we include the IF-Eval score as a continuous control variable. This ensures that the estimated effects of model size and training stage on MMLU-Pro are not confounded by a model's ability to follow instructions.

We estimate five nested OLS regression models: 1. Baseline size, 2. Adding training stage, 3. Adding instruction-following, 4. Swapping instruction-following for size and training stage interaction, 5. Adding instruction-following back in for the full model.

This progressive approach assesses robustness and reveals how the performance premium is driven by the underlying combination of model scale and training methodology.

Results

Descriptive Statistics

Table 2: Mean Descriptive Statistics by Training Stage

Training Stage	Count	Size- Billion Parameters	Size Range	MMLU-Pro	IfEval
Base-Merge	1708	10.700976	0.007 - 72.706	30.30658	51.11609
Fine-Tuned	1715	9.960755	0.124 - 140.621	22.73263	41.66846
Chat-Instruct	618	15.973427	0.033 - 122.61	27.13665	54.05201
Pretrained	188	7.130101	0.002 - 72.706	11.94576	23.00576
Other	56	8.302643	0.031 - 73.406	17.09921	32.27783

The majority of models from the leaderboard have gone past pretraining to merge with other models, be fine-tuned based on domain data, or trained for chat interaction. On average, the largest models are chat, the best performing are merged, and the best instruction-following are chat/merge. The log-linear size and performance relationship is apparent (Figure 3, link).

Regression Results

Table 3 (link) includes the results of our models. Each model's coefficients are set at a size of 1B parameters and with Base-Merge as the reference training stage. In models 3 and 5, IFEval score is held at 0 (in reality, a rare occurrence). Model 1 shows a 19 point performance increase for a 10x increase in size. 1B to 10B and then 10B to 100B each result in the same 19 point increase which illustrates the scaling law diminishing returns. Wald tests were conducted with each model iteration shown in Table 4 (link), showing that each one added value over the previous version.

Model 2 adds Training Stage, showing that merged models scale best with a 13 point performance jump per 10x size increase while Chat-Instruct and Fine-Tuned add 10 points and Pretrained lags behind with 6. Pretrained models get half the performance benefit from scale that merged models get.

Adding instruction-following score (Model 3) demonstrates a positive relationship between performance and ability to follow instructions. For every 10 points of IFEval score, a model gains 3 points of performance on average. This 0.29 coefficient is modest, which is expected given that both these benchmarks are measured on the same scale. Worth noting is that the R^2 value increases from 0.478 to 0.619, signaling that IFEval accounts for more variance.

Model 4 swaps instruction-following for the interaction between size and training stage. The R squared value drops back down without IFEval. The model shows that at 1B parameters, different training stages are close together: clustered around 8-10. Then with scaling, a 10x size increase yields 21 points for Base-Merges, 20 for Chat-Instruct, and 19 for Fine-Tuned, but that Pretrained lags behind at only 10.

Our full model adds instruction-following back in for similar results. The R^2 value goes up to 0.625. At 1B and 0 IFEval a Base-Merge starts at -1, Fine-Tuned 0, Chat-Instruct at -3, and Pretrained 2. Going from 1B to 10B adds 17 points of performance score for Base-Merges, 15 for Chat-Instruct, 14 for Fine-Tuned, and 9 for Pretrained. Overall, some training stages benefit from scaling more than others, and Pretrained models benefit the least. A Pretrained model with median instruction-following score outperforms other training stages at 1B parameters with a performance score around 15, but at 10B is eclipsed by the other stages (with a merged model scoring 29 and Pretrained 24).

The residuals vs fitted values plot Figure 4 (link) shows that prediction is less reliable for the smallest models (high residuals due to predicting impossible negative scores). The average residual

line hovers around zero once fitted values become positive. The benchmark score thresholds are visible in the bands at the top and bottom of the distribution.

Practical Sweet Spot

The plot of our full model's predicted values across size by training stage, set at the median instruction-following score, tells a compelling story.

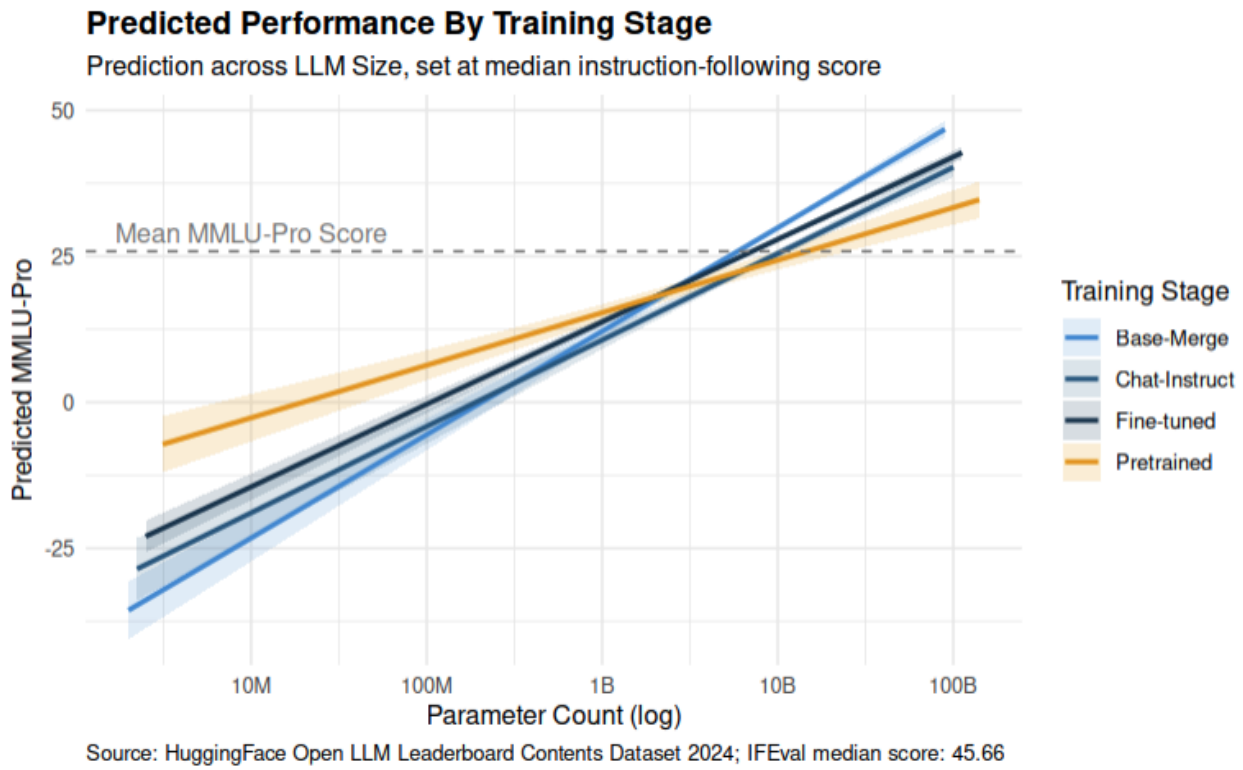


Figure 5: Predicted values with scale by training stage

There appears to be a "sweet spot" of size and performance tradeoff around 10B parameters where a model has average or above average general performance but is still on the smaller side. At that level, Base-Merged models perform better on average and are the wiser choice. The model doesn't predict well on the very small end of the scale, predicting negative values. If an organization only has capacity for a 1B parameter model or smaller, it makes sense to stick with a Pretrained model. That difference disappears at around 5B as performance improves for all training stages. Then at 10B Base-Merges tend to perform better but every stage is at or above the average performance level. The gain from 10B to 100B is the same as 1B to 10B within each individual training stage, so if budget and size are constraints, it doesn't make much sense to pursue marginal gains past 10B.

Discussion and Conclusion

Our analysis provides clear evidence that organizations with limited resources do not need to rely on massive and expensive proprietary models to achieve reliable AI performance. Nonprofits operating under strict budget and hardware limitations can make informed choices knowing the different return on investment for scaling up across training stages.

Organizations with the strictest hardware constraints will find that models under 5 billion offer limited performance on average. Here Pretrained models actually perform marginally better than their counterparts and are on average a better choice among limited options.

As hardware capacity allows for slightly larger models a clear sweet spot emerges around the 10 billion parameter mark. Models of this size can still be run on hardware as small as a laptop. Base-Merge models drastically separate themselves from the rest of the open source ecosystem at this stage. Our full interaction model highlights this divergence: at 10B parameters, Base-Merge models offer a transformative leap in performance capability without requiring enterprise level computing power.

Our analysis also uncovered the relevance of instruction-following ability in its positive relationship with general performance and potential tradeoff with reasoning. Better instruction-following is associated with better general performance, but excellent instruction-following can come with disproportionately low reasoning ability and vice versa. Organizations should understand tradeoffs and balance these based on the purpose of their tool.

The barrier to entry for high quality artificial intelligence is ultimately lower than it appears. Organizations can maximize their technological impact while remaining firmly within their operational constraints by strategically targeting the 10 billion parameter range and considering Base-Merge models. This knowledge allows organizations to narrow their search from thousands of potential models to a manageable group with some assurance that they are getting an efficient amount of performance for size.

Limitations and Future Research

Key Limitations

Our analysis is based on the Open LLM Leaderboard, which means it primarily covers open-weight models rather than the full landscape of closed, frontier systems. Many proprietary models do not share basic details such as parameter counts, training setups, or fully comparable benchmark scores, so we cannot draw clean conclusions between open and closed models. In addition, the data is time-bounded: it includes results up to March 2025, when the leaderboard was retired, so our findings reflect a specific historical snapshot rather than a current view. Individual models submitted to the leaderboard can also share similar architectures and base models as others,

or be submitted at various training stages. The modeling training stage partly addresses the shared variance, but not completely.

Finally, our measure of performance comes from benchmark scores. These metrics may not fully capture real-world tasks, must change over time as models are trained specifically to do better on them, and they ignore other key determinants of performance such as data quality, alignment and safety procedures, and architectural differences.

Future Research

Future research can extend this work in several directions. As new leaderboards and benchmark suites appear, the same capacity–performance framework can be re-run on fresher data to see whether the relationships we observe here still hold for newer model generations and task distributions. Where more transparent information on proprietary systems becomes available, those models could be folded into the analysis, allowing for more direct comparisons between open and closed offerings.

Later studies could move beyond parameter count and training stage by incorporating richer model descriptors such as architecture family, base model, alignment method, or fine-tuning strategy. For nonprofits and other budget-constrained organizations, a particularly valuable next step is to pair benchmark-based reasoning scores with practical deployment constraints; cost, latency, and hardware requirements; to identify models that are not just strong on paper, but also realistically affordable and deployable in day-to-day work.

Figures and Tables

Data and Methods

Figure 1: Benchmark Correlation

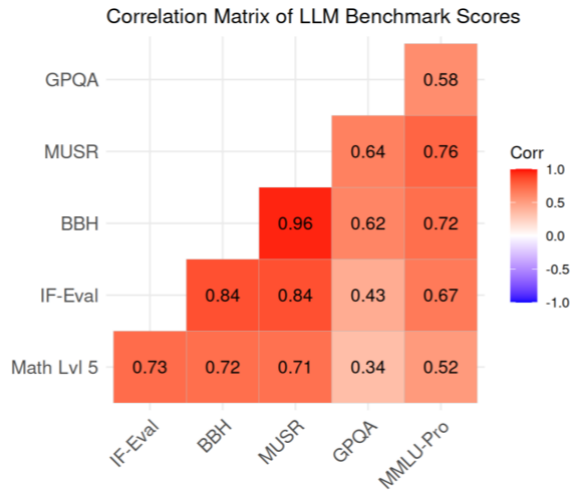


Table 1: PCA Loadings

	PC1	**PC2**
Proportion of Variance	0.735	0.125
if_eval	0.327	-0.757
bbh	0.455	0.0341
math_lvl_5	0.394	-0.229
gpqa	0.424	0.3111
musr	0.375	0.525
mmlu_pro	0.459	-0.014

Regression Results

Figure 3: Scaling Laws - Size and Performance

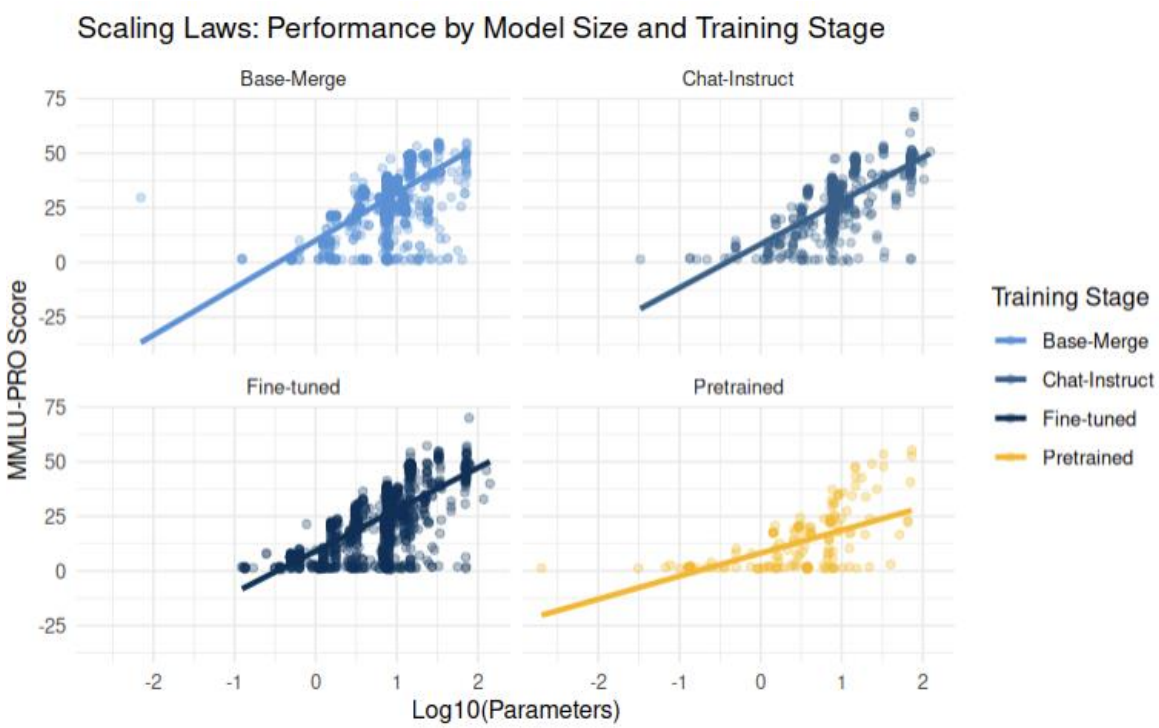


Table 3: Performance with Scale by Training Stage

Model Performance Across Size, Training Stage, and Instruction Following

	Dependent variable:				
	MMLU-PRO Score				
	Simple	Main Effects	+IFEval	Interaction	Full Model
	(1)	(2)	(3)	(4)	(5)
Log(Parameters)	19.554***	18.472***	14.066***	21.682***	17.712***
	(0.383)	(0.415)	(0.429)	(1.616)	(1.422)
Chat-Instruct		-3.421***	-4.236***	-1.785	-1.514
		(0.451)	(0.377)	(1.737)	(1.530)
Fine-tuned		-3.583***	-1.719***	-1.015	1.602
		(0.378)	(0.329)	(1.537)	(1.309)
Other		-6.741***	-2.669***	-0.959	2.855
		(1.185)	(0.853)	(2.183)	(1.774)
Pretrained		-7.819***	-1.956**	-1.946	3.168**
		(0.957)	(0.846)	(1.654)	(1.416)
IFEval Score			0.298***		0.294***
			(0.010)		(0.010)
Log(Params) × Chat-Instruct				-1.775	-2.916*
				(1.841)	(1.597)
Log(Params) × Fine-tuned				-2.614	-3.585**
				(1.666)	(1.424)
Log(Params) × Other				-7.991***	-7.415***
				(2.835)	(2.212)
Log(Params) × Pretrained				-11.158***	-8.709***
				(2.020)	(1.786)
Constant	9.830***	13.074***	1.951***	10.080***	-1.249
	(0.318)	(0.450)	(0.499)	(1.515)	(1.289)
Wald F-test		38.658***	863.272***	13.997***	826.563***
Observations	4,285	4,285	4,285	4,285	4,285
R^2	0.457	0.478	0.619	0.490	0.625
Adjusted R^2	0.457	0.478	0.618	0.489	0.624
Residual Std. Error	10.458 (df = 4283)	10.257 (df = 4279)	8.772 (df = 4278)	10.149 (df = 4275)	8.703 (df = 4274)

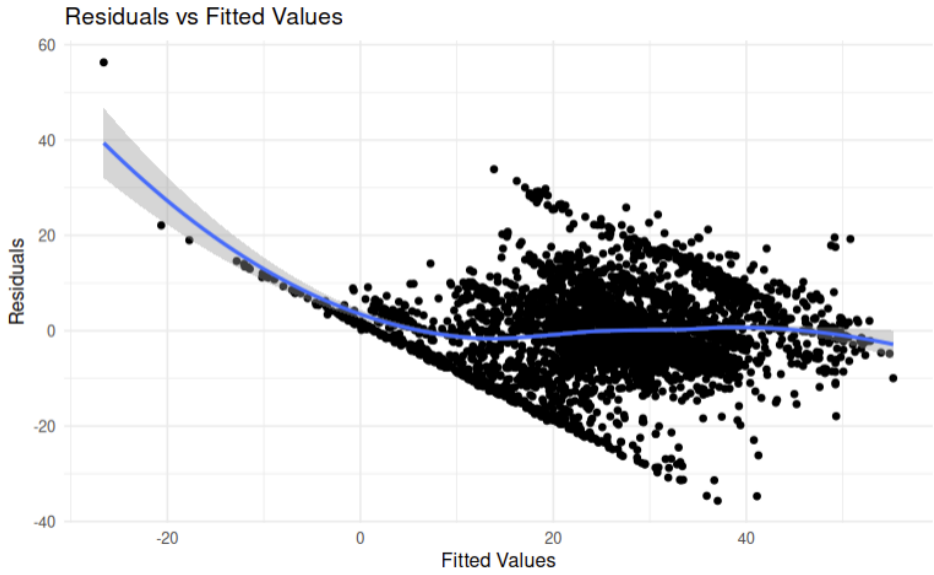
Note: *p<0.1;* ***p<0.05;*** p<0.01

Note: 'Base-Merge' is the reference category for Training Stage. Robust standard errors (HC3) reported in parentheses.

Table 4: Model Improvement with Iteration

Comparison	**F Statistic**	**p-value**	**df**	**df-residual**
simple vs model_2	38.90	4.86e-32***	4	4279
model_2 vs model_3	860.89	1.416e-172***	1	4278
model_2 vs model_4	16.37	2.58e-13***	4	4275
model_3 vs full	9.46	1.31e-07***	4	4274
model_4 vs full	822.99	1.11e-165***	1	4274
simple vs full	131.50	2.56e-219***	9	4274

Figure 4: Residuals vs Fitted Values



Appendix

Data Source

HuggingFace Open LLM Leaderboard: https://huggingface.co/datasets/open-llm-leaderboard/contents

Benchmarks (normalized accuracy-style percentage points)

Benchmark	**Definitions**
if_eval	Measures a model's ability to follow specific, verifiable instructions
Bbh	Evaluates complex, multi-step logical reasoning and planning
Math_lvl_5	High-school competition-level math problems. Level 5 is the hardest category in the original MATH dataset
gpqa	Extremely difficult multiple choice questions written by experts in Biology, Physics, and Chemistry. Designed to be "Google proof,": not answered via web search
musr	Complex, long-range reasoning over narratives
Mmlu_pro	Challenging, reasoning-focused questions across diverse disciplines

Variable Construction

Dependent Variable: MMLU-Pro benchmark score (mmlu_pro)

Key Independent Variable: Base-10 logarithm of **model size** (log_size), defined as log10 of parameter count. This transformation reduces right-skewness in size and lets coefficients be

interpreted as approximate changes in MMLU-Pro score associated with tenfold changes in model size. Reference at 1B

Control Variables: IF-Eval score (if_eval), reference at zero; **Training stage** (training_stage), reference is base-model; **Interaction** between **log10(size)** and **training_stage** (**log_size * training_stage**)

Model Specifications Attempted

No additional models were attempted in order to stay within our original constraints and mitigate deceptively small p values that could result from our exploration of the data. Architecture and base model variables were considered and could be pursued in the future. Both would require extensive parsing. Too granular or too broad would not reveal meaningful categories and data would be further segmented into smaller groups. We considered including CO2 emissions as a variable since sustainability is relevant for nonprofit organizations but ultimately decided against it due to a lack of consistent data.

Model Assumptions

Our large sample of 4,285 models allows us to rely on the asymptotic properties of ordinary least squares estimators. Given this sample size, we evaluated the large sample assumptions: (1) IID data and (2) that a unique Best Linear Predictor (BLP) exists. We observed no perfect multicollinearity as all variables have well defined coefficients. Regarding independence, models can be submitted at various training stages, share architecture, or share a base model. While we mitigated this by controlling for the training stage and deduplicating by precision, we acknowledge that strict independence is violated. As a result, we assume our standard errors may be deceptively low and conservatively trust only the most highly significant p-values.

While "fat tails" in our Q-Q plot indicate departure from normality at the extremes, our large sample size allows us to rely on asymptotic normality without biasing our estimators. HC3 robust standard errors were used for potential heteroskedasticity.

References

- https://huggingface.co/spaces/open-llm-leaderboard/blog
- Kaplan, Jared, et al. "Scaling Laws for Neural Language Models." arXiv.Org, 23 Jan. 2020, arxiv.org/abs/2001.08361.
- Hoffmann, Jordan, et al. "Training Compute-Optimal Large Language Models." arXiv.Org, 29 Mar. 2022, arxiv.org/abs/2203.15556.